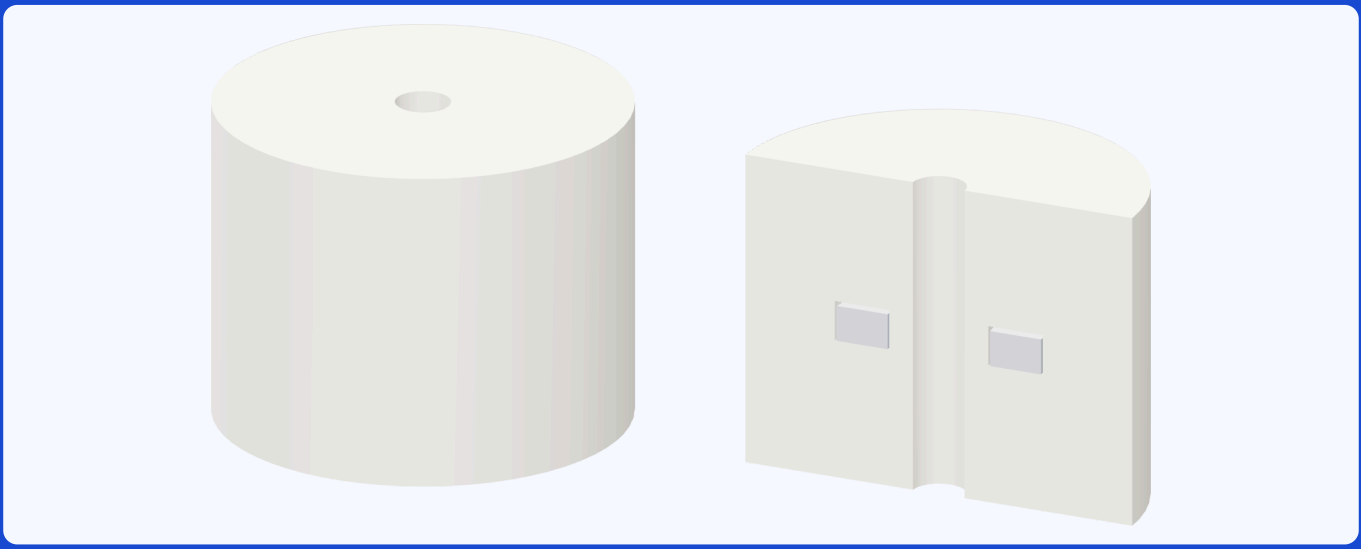
HOME SODA MACHINE

Magnetic float

One ASA Aero body. One ring magnet.
Pause, seat the ring, and print the pocket closed.

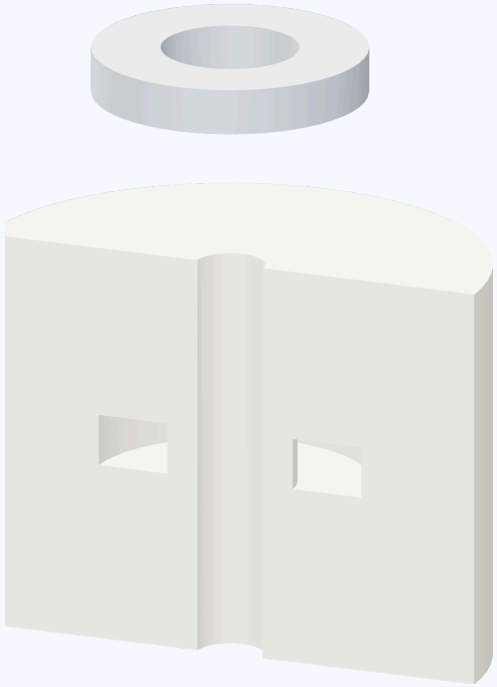


ASA Aero White · RC62 · 36 × 28 mm

Illustrated build guide · 13 pages

One body. One ring.

The printer makes one connected foamed body. A pause lets the ring enter its annular pocket before the covering roads close it.



The ASA Aero body and purchased RC62 ring are the complete assembly. The exploded view exposes the internal pocket; the covering body prints in the same job.

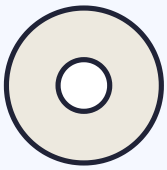
Float body	36 × 28 mm · ASA Aero throughout
Open guide bore	4.8 mm on a 3.175 mm rod
Ring magnet	RC62 · 19.05 × 9.525 × 3.175 mm
Magnet center	14 mm above the bottom
Current pocket	3.60 mm deep · CAD v2-upper-clearance

THREE ARTICLES

The carbonator and both flavor reservoirs each use one of these floats. The central bore stays open; the ring pocket is enclosed.

Dry one Aero spool

Use the selected ASA Aero White stock. Complete drying before loading the external feed.



SUNLU E2

80 °C · 8 hours

Forced-air drying

POLYMAKER DRYBOX

Sealed · below 20% RH

External right feed

Drying temperature

80 °C

SUNLU E2 · forced air

Drying time

8 hours

Before transfer to the drybox

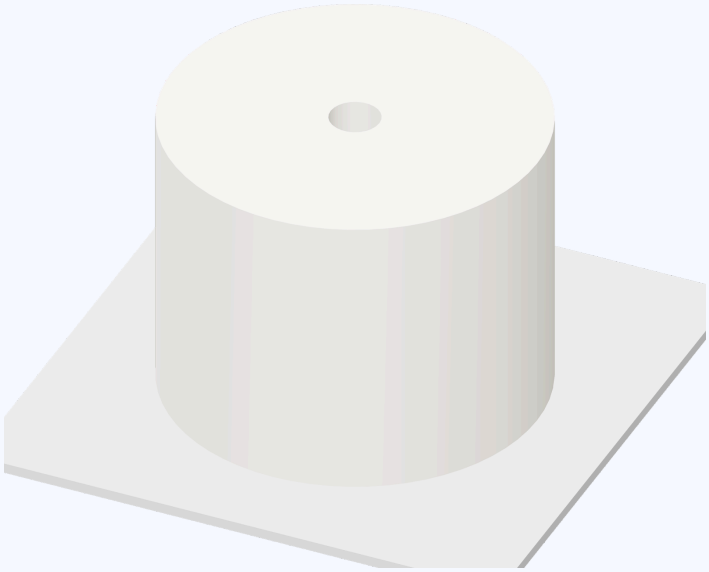
1. Dry **Bambu ASA Aero White (GFB02 / 46100)** in the SUNLU E2 for the complete cycle.
2. Transfer it to the sealed **Polymaker drybox**. Maintain **less than 20% RH** during storage and feeding.
3. Connect that drybox to the printer's external right feed. Keep the filament dry through the print.

FILAMENT PREPARATION

These are the ASA Aero manufacturer's forced-air drying conditions. They apply to filament before printing; the finished float and its magnet receive no drying bake.

Set up Mark2

The selected H2C recipe uses its fixed right nozzle and a glued Engineering plate.



One upright body on a cropped Engineering plate. The central bore is vertical and the bottom lies directly on the bed.

Right nozzle	right fixed hardened standard-flow 0.4 mm
Filament feed	Polymaker drybox · external right slot 255
Send tile	**Ext ASA-AERO R**
Plate / trim	Engineering, Bambu liquid glue · +0.04 mm user trim

1. With the printer cool, confirm the **right hardened standard-flow 0.4 mm nozzle** and matching printer setup.
2. Wash the Engineering plate with dish detergent and warm water. Rinse, dry, and apply a thin even coat of the owned Bambu liquid glue.
3. Set one RC62 and a soft brush beside the printer. Keep the enclosure closed during printing and run the shop ventilation.

RECIPE SCOPE

The recorded recipe is a trial process. It does not establish foam density, completed buoyancy or pressure life.

Slice the current body

Use the current v2 body and review its native slice before any submission.

Current geometry	float-aero.stl all_aero_float.rs
Nozzle / bed / chamber	270 / 90 / 60 °C
Flow / layer / line	0.52 / 0.2 mm / 0.48 mm
Walls	300 loops · nested foaming perimeters
Other paths	0 sparse infill, top/bottom skin, skirt, brim and support

1. Open the current STL with the retained right-nozzle ASA Aero process. Keep the body upright on the plate.
2. Use prepare_print.rs to prepare a separate v2 project and review record. It does not submit a print.
3. Review the actual native geometry, recipe, pause and covering roads. Bind the reviewed project/archive to the current source hashes before submission.

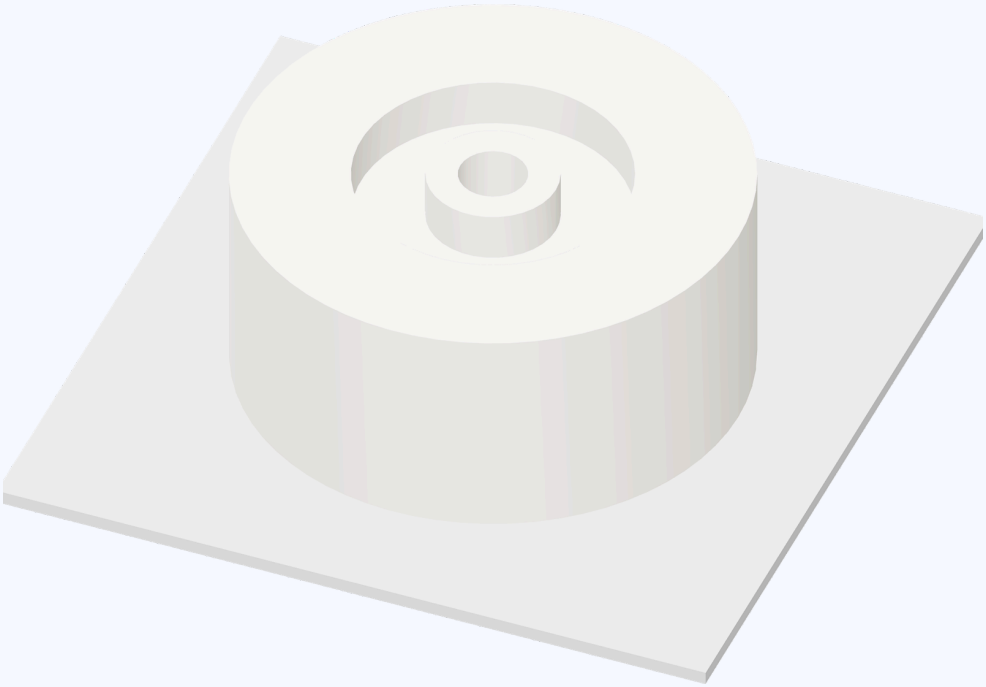
CURRENT PREPARATION STATE

The current CAD has a **3.60 mm pocket**. Its v2 native archive has not been prepared or submitted. The saved **all-aero-float.3mf** belongs to the identified v1 article and is listed as evidence on page 13.

Nested perimeters include seams, wipes and short transitions. This process is not a zero-travel spiral.

Keep the pocket open

The pause must occur before the first extrusion that covers the ring pocket.



Current v2 geometry at the expected last open plane. The annular pocket receives the RC62 while the body stays attached to the plate.

Pocket floor / roof	Z12.4125 / Z16.0125 mm
Expected last open layer	Z16 mm
Expected first covering layer	Layer 81 · Z16.2 mm

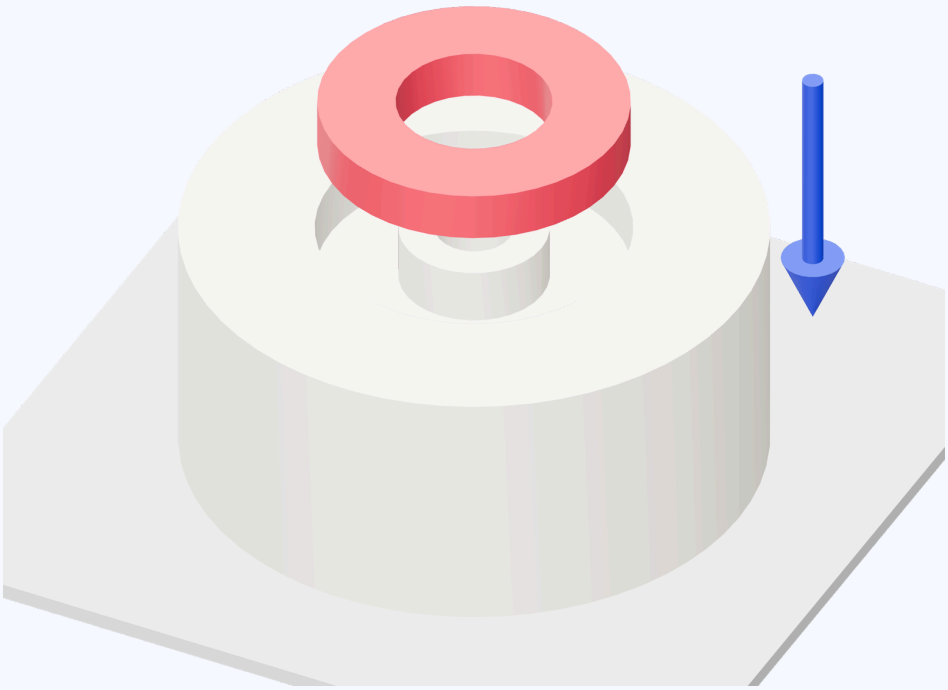
1. Inspect the **seat, last open pocket, pause command and first covering roads** in the new native slice.
2. Confirm one insertion pause occurs before any covering extrusion. The guide bore must remain open.
3. Confirm the seated ring lies below both pocket rims and the covering roads land on the inner collar and outer band.

EXPECTED, NOT A REVIEWED ARCHIVE

The layer-81 / Z16.2 position follows the current CAD and 0.2 mm layer grid. A fresh native slice must verify the emitted moves; a CAD section alone does not authorize a print.

Seat one RC62

The printed body stays on the Engineering plate. Put only the ring magnet into the open pocket.



Lower the RC62 along the guide axis into the open annular pocket. Coral identifies the ring being added; the arrow shows its motion.

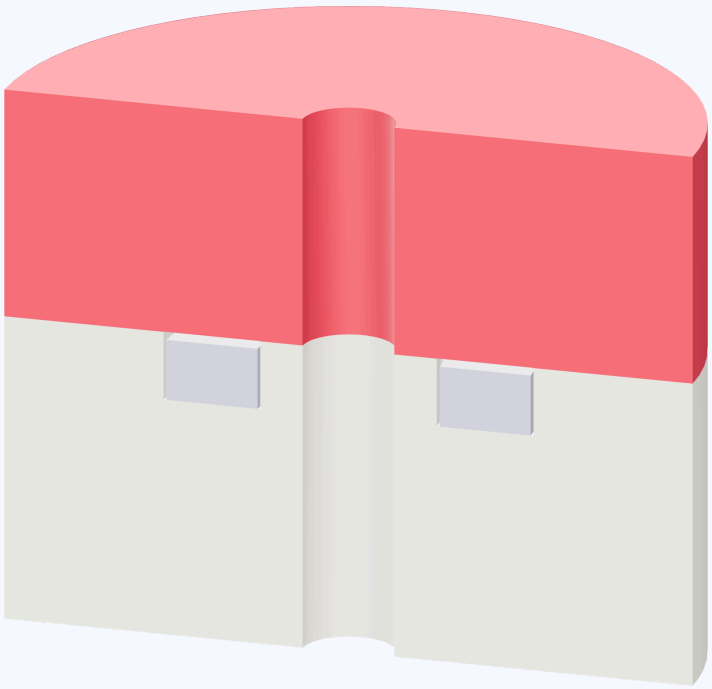
1. Wait for the programmed pause and toolhead parking before reaching into the printer.
2. Leave the sheet mounted and the float attached. Separate **one RC62** from the stack and lower it squarely into the pocket.
3. Seat it fully on the pocket floor with its hole coaxial with the guide bore. Its flat faces lie horizontal, **below both pocket rims**.
4. Clear loose strings without shifting the body. Keep the central guide bore open.

RESUME CONDITION

The ring sits flat and fully down. Nothing projects into the covering path, and no hand or tool remains inside the printing area.

Resume over the ring

The same ASA Aero job builds the material above the pocket and closes the ring inside the body.



Section through the current body. Coral shows the material printed above the pocket; the ring is centered at the body's axial midplane.

1. Remove hands, tools and loose strings from the printer. Close the enclosure.
2. Manually resume the same job. Let the printer restore its temperatures and continue the covering roads.
3. Observe the initial covering layers for ring movement, dragging or lifting. Record the result with that article.

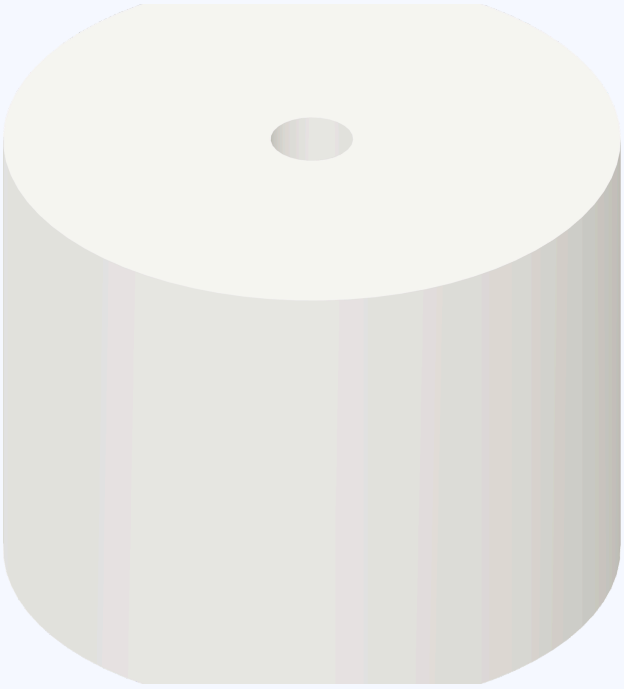
WHAT THE EXISTING PRINT DEMONSTRATES

The identified v1 article supports paused insertion and the initial covering deposition. It does not establish the finished roof or retained signal for a cooled float.

RC62 has an 80 °C continuous operating limit. The 270 °C nozzle setting does not tell us the magnet's actual temperature; retained field after insertion is unmeasured.

Release and inspect

Let the print cool before removing it. Keep the foamed body and its open bore intact.



The complete 36 × 28 mm Aero body. The ring lies at Z14; the through-bore remains open.

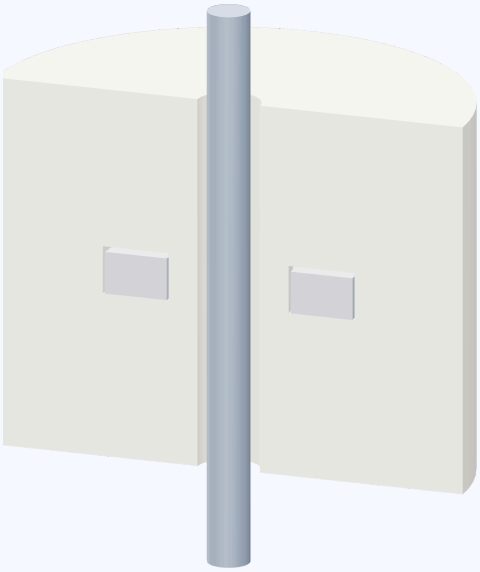
1. Leave the enclosure closed for initial cooling. Use the shop handling threshold of **35 °C or lower bed temperature** before releasing the part.
2. Lift the Engineering sheet and flex it gently. Support the float rather than prying through its bore or pocket roof.
3. Clip loose strings and brush away crumbs. Preserve the bore, pocket closure and cylindrical outside surface.
4. Inspect the covering area and slide the finished float on its actual rod. Record the finished article's condition.

PHYSICAL SCOPE

Finished roof integrity, cooled removal, retained magnetic signal and sliding have no acceptance record. The cooling threshold is a conservative shop handling choice, not an ASA-specific manufacturer release limit.

Locate the guide rod

Each 3.175 mm rod stands 20 mm inward from the vessel's wet inside wall. Use the vessel datum, not the end-cap edge.



An explanatory 3.175 mm rod through the actual 4.8 mm float bore. This picture shows the guide interface, not measured liquid performance.

Rod / radial running play	3.175 / 0.8125 mm
Body-to-wall clearance	2 mm nominal · 1.1875-2.8125 mm upright range
Carbonator finished rod	5.110 ±0.005 in (129.79 ±0.13 mm) · FR-CARB-01 finished length and end form
Reservoir rod cut	203.750 mm · 0.15 mm axial clearance
Greatest float-edge to reed distance	5.713 mm carbonator / 11.062 mm reservoir

Use the installation record for the carbonator blind-drill coordinates and the matching reservoir body/cap bosses. Confirm free motion over the actual travel.

CARBONATOR CAPTURE

The 36 mm float is captured on the rod before final vessel closure. It cannot pass through the NPT ports. Welding exposure and post-closure function have no acceptance record.

Calibrate the crossings

The magnet midplane is not the liquid surface. Final reed heights come from the finished float in the actual liquid and wall fixture.

Guide drilling datum

20 mm

Rod axis from the wet inside wall

Reed design maximum

18 mm

Nearest float edge to reed center, including guide play

1. Use the actual rod, finished float, MDSR-7-10-15 reeds, vessel wall, ruler or calipers, multimeter and existing measuring cup.
2. In water and actual flavoring, record **a = magnet center minus liquid surface height**. Magnet center is float bottom plus 14 mm.
3. Sweep slowly upward and downward past each reed. Record every closure and release as **d = magnet center minus reed center**, including separate activation lobes.
4. Place each reed at **target liquid height + a - d**, using the closure in its operating direction. Repeat three cycles at nominal position and greatest permitted retreat.

PLACEMENT ACCEPTANCE

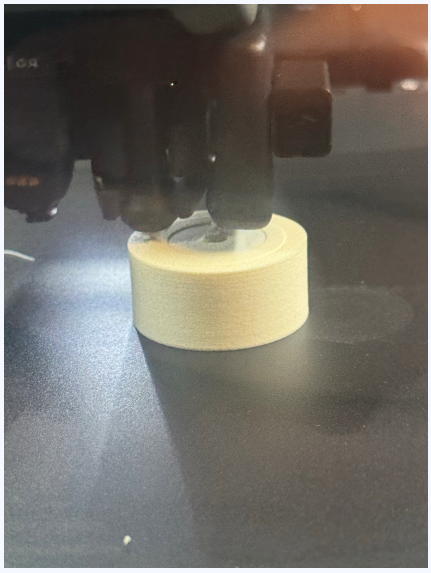
Intended crossings must occur in order before the travel stops, with no missed closure or unintended trigger. Repeated liquid crossing heights must span at most 2 mm. Use measured liquid volumes for reservoir quarter marks.

Reeds pulse as the float passes. Reservoir control retains the last crossing and flow direction. Carbonator high inhibits refill even if low remains closed; a refill timeout still needs an explicit clear.

The calibration procedure defines the target levels and recording method. Those levels remain provisional until the liquid crossings are measured.

What is recorded

Mark2 task 1306080180 identifies the paused v1 article. Its pocket and emitted pause belong to that frozen source.



The operator photo partly obscures the covering roads. The report supports insertion and initial overprinting in this identified v1 trial.

v1 pocket	3.40 mm · frozen native source
v1 insertion / first cover	After layer 79 at Z15.8 / layer 80 at Z16.0 mm
v1 slice estimate	88 minutes · manual pause excluded
Operator result	Ring insertion worked; first covering layer was tight, second deposited well
Printed-float reed bench report	20 mm usable; 21-24 mm intermittent; 25 mm absent Datum: nearest float edge to reed center

LIMITS OF THESE RESULTS

Final cooled roof, density, buoyancy, retained signal, liquid uptake, pressure life and flavor compatibility are unreported. The reed bench report does not identify its reed, wall fixture or repeat count.

The current sizing model assumes 0.65 g/cm³ Aero: about 22.75 g assembled mass and 5.23 g spare lift. These are calculations, not measurements.

Files and recipe sources

Current geometry, print preparation and physical evidence are identified separately.

Repository files

Current body	CAD source for the one-piece ASA Aero float.
Current STL	Current v2 geometry; prepare and review a fresh native slice.
Design and installation	Dimensions, selected process and qualification limits.
Rod and reed datums	Drilling, guide motion and directional liquid calibration.
Print preparation	Writes a separate v2 project/review; submits no print.
Identified v1 project	Frozen v1 project, 3.40 mm pocket; evidence for task 1306080180.
Identified v1 review	Native paths, pause and source hashes for that article.
Physical observations	Insertion, initial overprint and reported reed reach, with their scope.

Manufacturer guidance

Bambu ASA Aero TDS	Filament drying, storage humidity and material descriptors.
Bambu ASA Aero guide	Foaming, plate selection and printing guidance.
K&J RC62	Ring dimensions, N42 grade and continuous operating limit.
Littelfuse reed activation	Closure, hold, release and orientation effects.

Prepared 3 October 2026. Illustrations derive from current v2 CAD. Warm white represents Aero and nickel the ring; coral marks the added ring or covering material. Cutaways, plate crops, rods and arrows explain geometry and motion.